

Single-Cell Colon Cancer Annotation and Classifier Analysis

1. Single-Cell Annotation

The raw dataset.h5ad dataset containing 5,561 cells × 14,016 genes was processed and annotated into specific cell types and microenvironment subpopulations. The annotated AnnData file contains the normalized matrix, UMAP embeddings, and obs['final_cell_type'] labels for downstream classifier analysis.

Cell Type	Number of Cells	Percentage
Epithelial Adenocarcinoma Cells	1,313	23.61%
Tcm / Naive Cytotoxic T cells	699	12.57%
Intestinal Macrophages	666	11.98%
Endothelial Cells	634	11.40%
Stromal Fibroblasts	553	9.94%
NK Cells	200	3.60%
B Cells	172	3.09%
Tcm / Naive Helper T cells	156	2.81%
Double-Positive Thymocytes	118	2.12%
Monocytes & Macrophage Subsets	234	4.21%
Effector & Resident T Cells	169	3.04%
Other Microenvironment Cells	648	11.63%
Total annotated cells	5,561	100%

Key Marker Gene Signatures

Cell Type	Key Marker Gene Signatures
Epithelial Adenocarcinoma Cells	EPCAM, KRT8, KRT18, CDX2, MUC2
Tcm / Naive Cytotoxic T cells	CD8A, CD8B, LEF1, TCF7, CCR7
Intestinal Macrophages	CD14, CD68, FCGR3A, CSF1R
Endothelial Cells	PECAM1, VWF, CD31
Stromal Fibroblasts	COL1A1, COL1A2, ACTA2, PDGFRA
NK Cells	NKG7, GNLY, NCAM1
B Cells	MS4A1 (CD20), CD19, CD79A
Tcm / Naive Helper T cells	CD4, IL7R, CCR7
Double-Positive Thymocytes	CD4, CD8A, PTCRA
Monocytes & Macrophage Subsets	CD14, FCGR3A, S100A8, S100A9
Effector & Resident T Cells	Th1 (IFNG), Th17 (RORC), Tem/Trm
Other Microenvironment Cells	Plasma cells, Mast cells, Dendritic cells

2. Classifier Analysis

The annotated h5ad dataset was prepared for downstream classifier analysis using the final_cell_type labels. The purpose of the benchmark was to compare classification using the complete feature set with classification using the top 500 FGL-SAN biomarker features.

Validation strategy: 5 repetitions × 10-fold stratified cross-validation = 50 evaluation runs.

Evaluation Metric	Full Dataset	Top 500 Biomarkers
Accuracy	39.62% ± 0.9292	18.78% ± 0.7552
Precision	38.67% ± 0.9445	10.38% ± 6.3181
Recall	39.62% ± 0.9292	18.78% ± 0.7552
F1 Score	38.97% ± 0.8650	10.84% ± 0.3306
MCC	0.3391 ± 0.000108	0.0754 ± 0.000127
Cohen's Kappa	0.3389 ± 0.000108	0.0669 ± 0.000100
AU-ROC	0.8443 ± 0.000021	0.6196 ± 0.000058
AU-PRC	0.3935 ± 0.000086	0.1350 ± 0.000031

3. Generated Analysis Files

File	Description
annotated_colon_cancer.h5ad	Annotated single-cell dataset with normalized matrix, UMAP embeddings, and final_cell_type labels
annotate_colon_cancer.py	Standalone Python cell annotation code
colon_cancer_cell_distribution.csv	Cell-type abundance table
colon_cancer_differential_markers.csv	Top 50 ranked marker genes per cell type with log2FC and adjusted p-values
dataset.h5ad	Raw input dataset: 5,561 cells × 14,016 genes

4. Summary

The colon cancer single-cell dataset containing 5,561 cells was annotated into 12 major cell types/subpopulations. The annotated dataset was then used for downstream classifier benchmarking. Across 50 repeated stratified cross-validation runs, the full feature dataset performed better than the top-500 FGL-SAN biomarker dataset across all eight reported metrics. For example, AU-ROC was 0.8443 for the full dataset compared with 0.6196 for the top-500 set, while F1 score was 0.3897 compared with 0.1084.

Note: The classifier algorithm itself is not specified in the information provided here.